

BEV Navigator — User Journey · Concept draft

For discussion on 2 Oct 2026 · Zurich × HSG Consultancy Project · neutral decision support for switching from a combustion car to a battery-electric vehicle (BEV)

Read left → right. First understand the person (short profile → persona → targeted questions), then answer the core question: does switching pay off? Options and info boxes show how to make it work. Neutral: no provider or insurer offers.

1 · Quick profile

Round 1 · ~5 clicks · ~1 min

Start · user opens the BEV Navigator (chat or click)

Buttons & chips, no typing. "Charging" is not mentioned yet.

ASKED

Q1 · Current car · class · fuel · age — model optional.
Small · Compact · Mid / estate · SUV · Van

DERIVED · DB 1 · Consumption, resale value, costs of today's car · already a BEV → other path

ASKED

Q2 · Main use (multi-select)
Commute · Everyday · Leisure · Long trips · Holidays · Towing · Business

Opens or skips the round-2 modules

ASKED

Q3 · Distance per year
< 10,000 km · 10–20,000 · > 20,000 · Not sure

"Not sure" → persona default

ASKED

Q4 · Home & parking · house / apartment · rent / own · where the car sleeps

→ charging access derived, never asked directly

ASKED

Q5 · Postcode

DERIVED · DB 3 · 5 · canton, electricity price, charging points nearby

AHA 1 · after postcode · DB 5

Map of nearby charging points. Challenges the "where would I even charge?" misconception before any barrier is discussed.

2 · Persona

routing, not a result

Persona assignment

Based on the round-1 answers, the best-fitting persona profile is selected from a predefined set (4–6, to be defined) or generated. It decides which deepening questions follow and which AHA moments are shown → to get to know the user better. It routes the dialogue; it never replaces the individual calculation.

PERSONA DEFAULTS · DB 2

Each persona carries default values — annual km, charging mix home / work / public, vehicle class, typical trips → the user confirms instead of answering.

CHECKPOINT · CLICK

"You look like [persona]: ~12,000 km, mostly [commuting], [compact] class. Correct?"
Looks right · Adjust km · Adjust car class

3 · Deepening

Round 2 · only relevant modules · two strands

The persona steers two strands: A which questions are asked, B which AHA moments are shown. Only relevant modules appear; answers override persona defaults. ~1–2 min.

A · Routed questions

IF COMMUTE · Commute & workplace · days · distance · charging at work (yes / no / don't know)

IF TRIPS / HOLIDAYS · Longer trips & holidays · how often · longest trip · destinations · winter

IF TOWING · Towing · what · weight (kg) · how often

IF OWN HOUSE · Home energy · PV on the roof · special tariff

ALWAYS · SHORT · Timing & horizon · when is a new car due · how long to keep it

ALWAYS · SHORT · Preferences · smaller car + rental days OK? · used OK? · budget

skipped modules are not shown

B · AHA moments

AHA · DB 1 · "Charging at work would cover [80] % of your weekly driving."
"Don't know" → employer inquiry (Q3)

AHA · DB 1 · 5 · "A compact BEV covers [94] % of your trips without a stop."

AHA · DB 1 · "[n] BEV classes can tow your [1,500] kg trailer."

AHA · DB 15 · "Your PV could cover about [40] % of your charging."

AHA · DB 1 · 4 · "Your current car costs you about CHF [1,400] a year in fuel."

→ preselects matching options (e.g. right-sizing)

FULL PROFILE · user data

Full profile = persona + confirmed defaults + round-2 answers
This profile — not the persona label — feeds the calculator and the preselection of options.

Not asked up front: concerns and emotional factors (range anxiety, battery life, cyber security ...) → info boxes on the results page. Keeps the entry short.

4 · Calculator mask

user data + model data

YOUR INPUTS & ASSUMPTIONS · MOCKUP · ILLUSTRATIVE VALUES

MUST · Base case: no charging in your garage today → "Charging in shared garage" is preselected. Details on the results page.

YOU & YOUR DRIVING · USER DATA (FULL PROFILE)

Postcode: 9000 St. Gallen · ANSWER · Parking: Shared garage, rented · ANSWER · Annual distance: 12,000 km · ANSWER

Commute: 3 days · 18 km · ANSWER · Longer trips: 2x / month · 250 km · PERSONA · Charging mix home / work / public: 0 / 0 / 100 % · DERIVED

CURRENT CAR VS. BEV BASE CASE · MODEL DATA (DB N)

	CURRENT CAR	BEV BASE CASE
Vehicle	Diesel compact, 2016 · ANSWER	Compact BEV — e.g. [A], [B], [C] · DB 1
Price / resale value	Resale CHF 8,500 · DB 1	CHF 22,600 · range 19–27k · DB 1
Consumption	5.6 l / 100 km · DB 1	17 kWh / 100 km · DB 1
Energy price	2.10 CHF/l (was 1.95) · OVERRIDE	0.55 CHF/kWh public · DB 6
Insurance / year	CHF 900 · DB 9	CHF 950 · DB 9
Vehicle tax / year	CHF 380 · DB 8	CHF 120 · DB 8
Maintenance & tyres	CHF 1,100 · DB 7	CHF 650 · DB 7

SETTINGS

Horizon: 8 years · SETTING · Interest rate: 2 % · SETTING · Show ranges: low / base / high · SETTING

Every field shows its source and can be overwritten — down to a concrete offer (price, range).
Reset · Recalculate

BEV base case = same vehicle class as today + charging as possible today — no extra investment, no tricks. Not feasible → warning and the required option is preselected (MUST). Options already in place or wished are preselected too.

Data layer

Pre-fills every value in the calculator mask · referenced as "DB n" · sources = first candidates, to verify (access, licence, API)

MVP	DB 1 Vehicle archetypes BEV + combustion · 4–8 classes · low / base / high · 3 example models · resale by age	DB 2 Persona library Personas with default values	DB 3 Electricity price Per municipality via postcode · cand.: EICom	DB 4 Fuel prices Petrol / diesel, average	DB 5 Charging points CH Locations, AC/DC · cand.: BFE open data	DB 6 Public charging tariffs Ranges per type (AC / DC)
	DB 7 Maintenance & tyres Per class × drivetrain · cand.: TCS	DB 8 Vehicle tax — simplified Ranges / BEV reduction per canton	DB 9 Insurance Cost ranges: house · neutral approach to agree with Zurich	DB 10 Wallbox & installation Cost ranges: house · shared garage	DB 11 Rental car prices For rental days (right-sizing)	DB 12 Content library Info boxes, templates, to-dos
ADD-ON	DB 13 Vehicle tax — 26 cantonal formulas Exact tax old vs. new	DB 14 Subsidies Cantonal / municipal	DB 15 PV yield & self-consumption For PV + BEV charging	DB 16 Special tariffs Night / e-mobility tariffs	DB 17 Charging abroad Coverage per country / route	DB 18 Leasing conditions Criteria / typical terms
	DB 19 CO ₂ lifecycle Combustion vs. BEV lifecycle	SETTINGS · LISTED, NOT YET DEFINED Interest rate · horizon · ranges low / base / high · range factors (WLP → real, winter, reserve) · charging losses · energy price trend · battery ageing · residual value BEV vs. combustion · thresholds for "recommended"				

Legend	Question block — asked by click	Derived / persona default	AHA moment
Start	Note	Mockup — illustrative values	DB n · Uses database n
Processing	ADD-ON — to agree with client	MUST · Base case not feasible — explained	OVERRIDE · Value changed by the user
MVP — option or database			

5 · Results page — home base

calculation · 3 key figures · options · info boxes · my plan

YOUR BEV CHECK · MOCKUP · ILLUSTRATIVE VALUES

calculator

calculator mask — COLLAPSED (EXPANDED VIEW: STEP 4)

• Your inputs & assumptions · 16 values · 1 changed by you
12,000 km · ANSWER · Compact class · PERSONA · 0.55 CHF/kWh · DB · Diesel 2.10 CHF/l · OVERRIDE · 8 years · 2 % · SETTING · Edit

RESULTS

Switching pays off within your horizon — break-even in year 7, in year 6 with charging at work.

Cost per year
CHF 4,990
vs. CHF 5,390 today — 400

Break-even
Year 7
year 6 with recommended options

Cash needed at switch
CHF 16,000
price — resale value + one-off investments

Cumulative cost, CHF

Graph showing cash at switch, break-even, keep current car, recommended options, and BEV over 8 years.

OPTIONS — TICK TO INCLUDE, THE CALCULATION UPDATES INSTANTLY

PRESELECTED FOR YOU
Charging in your shared garage · MUST · No socket today → template letter to your landlord · +CHF 1,900 one-off — CHF 520 / year vs. public charging · Details >

Compact BEV + ~6 rental days a year · YOUR WISH · Right-sizing: smaller car, larger rental car for holidays · rental days included in the cost · Details >

RECOMMENDED FOR YOUR CASE
Charging at work · RECOMMENDED · Template inquiry to HR — CHF 350 / year · break-even ~1 year · Details >

Used BEV with battery-health check · RECOMMENDED · ~CHF 9,000 cash at switch — checklist: battery certificate, warranty, test · Details >

MORE OPTIONS
Leasing vs. buying · Details >

Switch timing — now or in 2 years? · Details >

Special electricity tariff · Details >

INFO BOXES — ANYTHING STILL HOLDING YOU BACK?

Battery life · Facts & sources >

Winter range · Facts & sources >

Cyber security · Facts & sources >

Charging abroad · Coverage map >

FINISH

Your next steps: write to landlord · ask employer · book a test drive
Finish & get my plan (PDF)

Three key figures — always shown

- Cost per year — annualised total cost incl. interest: status quo vs. BEV
- Break-even — the year from which the BEV is cheaper overall
- Cash needed at switch — purchase price — resale value + one-off investments of selected options

One chart shows all three: jump at year 0 = cash · slope = running costs · crossing = break-even. Plus a one-line verdict ("not yet" is a valid outcome).

Options — priced, change the calculation

01 Home charging — wallbox (own house) · MVP · DB 10	02 Charging in shared garage — tenant letter / STWE application · MVP · DB 10 · 12	03 Charging at work — employer inquiry · MVP · DB 12
04 Public charging plan / rented e-parking space · MVP · DB 5 · 6	05 Right-sizing — smaller BEV + rental days (2:1) · MVP · DB 1 · 11	06 Used BEV incl. battery-health check · MVP · DB 1
07 PV + BEV charging · ADD-ON · DB 15	08 Special electricity tariff · ADD-ON · DB 16	09 Subsidies · ADD-ON · DB 14
010 Leasing vs. buying · ADD-ON · DB 18	011 Switch timing · ADD-ON · DB 1	

Info boxes — qualitative, answer concerns

11 Concerns fact sheets — battery life, winter range, cyber security, residual value · MVP · DB 12	12 Charging on the road & abroad — coverage per country · ADD-ON · DB 17
13 Try before you buy — 48–72 h test, car sharing, BEV rental · ADD-ON · DB 12	14 CO ₂ comparison — combustion vs. BEV lifecycle · ADD-ON · DB 19

LEGEND — EVERY OPTION / INFO BOX

What it is · effect (one-off CHF · CHF / year · break-even shift — or info only) · to-do (next steps, template or link) · data (DB n)
PRESELECTED · in place / wished · MUST · base case not feasible · RECOMMENDED · pays off in this case (green, on top) · MORE · standard, below
Open list — new options plug in the same way. MVP / add-on split is a proposal to agree with Zurich.